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1 Examples

We begin with a minimal example that has non trivial higher differential:

Example 1.0.1 (The Sphere). If we take the three-sphere $S^3 \subseteq \mathbb{R}^4$ and denote the north pole with N and the south pole with S we get the Morse chain complex as

$$0 \longrightarrow \langle S \rangle_{\mathbb{Z}} \longrightarrow 0 \longrightarrow 0 \longrightarrow \langle N \rangle_{\mathbb{Z}} \longrightarrow 0$$

Then for the second page we get the Morse homology together with the 0 maps and on the third page we have:

$$\begin{array}{ccc} \langle S \rangle & \begin{array}{cc} 0 & 0 \end{array} & \langle N \rangle \\ & \searrow \quad \quad \nearrow & \\ & d_3^0 & \end{array}$$

Now we want to check what $d_3^0(S)$ is. For this we want to use its definition and hence we need to give the Morse filtration. For this we can use:

$$N_{-1} := \emptyset, \quad N_{0,1,2} := H_- \text{ the southern hemisphere}, \quad N_3 = S^3.$$

First we translate E_r into the language where the definition was formulated. We know that EM_3^p is naturally isomorphic to

$$E_3^{p,0} = \frac{\gamma^{-1}(\alpha^2 D^{p,-2})}{\beta(\ker(\alpha^2|_{D^{p+3-1,-2}}))}$$

Now for $r = 3$ and $p = 0$

$$\alpha^2 D^{2,-2} = \text{im}(K^0(N_2) \rightarrow K^0(N_0)) = \text{im}(K^0(H_-) \rightarrow K^0(H_-)) \cong \mathbb{Z}$$

Hence $E_3^{0,0} = \gamma^{-1}(\alpha^2 D^{2,-2})^{0,0} \cap E_0^{0,0} = K^0(N_0, \emptyset) \cong \mathbb{Z}$. And similar we conclude that $E_3^{3,0} = K^3(N_3, N_2) \cong \mathbb{Z}$. Now we analyze the map

$$d_3^{3,0} : K^3(N_3, N_2) \rightarrow K^0(N_0, \emptyset)$$

So far we were able to ignore all orientation, (which we formally need for the morse boundary operator). So let's fix the orientations of the unstable part by saying the unstable part of the north pole (i.e. the whole S^3) is the canonical orientation of $S^3 \subset \mathbb{R}^4$, whilst the unstable part of the south pole (i.e. the south pole) has the orientation -1 .

Now $d_r = \beta \circ \alpha^2 \circ \gamma$ and $\gamma : K^3(N_3, N_2) \rightarrow K^0(N_3)$ maps our generator induced by N to the standard generator of $K^3(S^3)$ induced by the chosen orientation, due to N_2 being a contractible subset. Now $\alpha^2 : K^3(N_3) \rightarrow K^3(N_1)$ is the zero map, since its range is 0 due to N_1 being homotopically equivalent to the point and 3 being odd. Finally $\beta : K^3(N_1) \rightarrow K^4(N_0, \emptyset)$ is the connecting homomorphism that is trivial here. Hence $d_3^{3,0}$ is the zero map. And thereby EM_4 which equals E_∞ is

$$\langle S \rangle \quad 0 \quad 0 \quad \langle N \rangle$$

Now how to we get information about our beloved K groups back? For this we remind ourself about the filtration

$$K^0(S^3) = F^{-1}K^0(S^3) \supseteq F^0K^0(S^3) \supseteq F^1K^0(S^3) \supseteq F^2K^0(S^3) \supseteq F^3K^0(S^3) = 0$$

Now using this filtration we have the short exact sequence

$$0 \longrightarrow F^0K^0(S^3) \longrightarrow K^0(S^3) \longrightarrow E_4^{0,0} = \frac{F^{-1}K^0(S^3)}{F^0K^0(S^3)} \longrightarrow 0$$

Hence, $K^0(S^3)$ is an extension of $F^0K^0(S^3)$ and $E_4^{0,0} \cong \mathbb{Z}$. Furthermore, we have the exact sequence

$$0 \longrightarrow F^1K^0(S^3) \longrightarrow F^0K^0(S^3) \longrightarrow E_4^{1,-1} = \frac{F^0K^0(S^3)}{F^1K^0(S^3)} = 0 \longrightarrow 0$$

Hence, $F^0K^0(S^3) \cong F^1K^0(S^3)$ and by the same argument we can continue this chain

$$F^0K^0(S^3) \cong F^1K^0(S^3) \cong F^2K^0(S^3) \cong F^3K^0(S^3) = 0$$

Hence, the first short exact sequence gives is an isomorphism between $K^0(S^3) \cong \mathbb{Z}$.

Now we want to calculate the K group of a non trivial example:

Example 1.0.2 (The Real Projective Space). Here, we define $\mathbb{RP}^n$ to be $\mathbb{R}^{n+1} \setminus 0$ modulo scalar multiples. Hence it is the space of lines in $\mathbb{R}^{n+1}$. Alternatively we can view it as the n -sphere where the operation from $\{\text{id}, -\text{id}\}$ is quotient out. From this description we get a metric, since $S^n \supseteq \mathbb{R}^{n+1}$ inherits the subspace metric in which all operations from $\{\text{id}, -\text{id}\}$ are isometries. Hence, this metric projects down to $\mathbb{RP}^n$. Now let

$$A = \text{diag}(\lambda_0, \lambda_1, \dots, \lambda_n)$$

be a symmetric matrix with real $\lambda_i \in \mathbb{R}$. Then we define the map

$$\Lambda : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{R}; \Lambda(x) := \frac{\langle x, Ax \rangle}{\|x\|^2}.$$

Notice that this function descends to a well defined function $\tilde{\Lambda} : \mathbb{RP}^n \rightarrow \mathbb{R}$. On the whole $\mathbb{R}^{n+1}$ we can define the function

$$f : \mathbb{R}^{n+1} \rightarrow \mathbb{R}; f(x) := \frac{1}{2} \langle x, Ax \rangle = \frac{1}{2} \sum_{i=0}^n \lambda_i x_i^2.$$

Now assume that $\varphi_t(x)$ is the flow associated to $-\text{grad}(f)$, meaning

$$\frac{d\varphi_t(x)}{dt} = -\text{grad}_{\varphi_t(x)}(f) = -A\varphi_t(x),$$

since

$$\frac{\partial f}{\partial x_i}(x) = \lambda_i x_i.$$

Now we have the lemma:

Lemma 1.0.3. *The flow lines of the negative gradient flow with respect to $\frac{1}{2}\tilde{\Lambda}$ on $\mathbb{RP}^n$ are the images under the quotient map*

$$\pi : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow \mathbb{RP}^n$$

of the flow lines of the linear system

$$\frac{d\Phi_t(x)}{dt} = -A\Phi_t(x)$$

Proof. Let x be a point on the flow line $\Phi_t(x)$. Then $-\text{grad}(f)(x) = -Ax$. Now we split this into a part that is orthogonal to y and one that is parallel:

$$-\text{grad}(f)(x) = \underbrace{-\Lambda(x)x}_{\text{parallel to } x} + \underbrace{-Ax + \Lambda(x)x}_{\text{orthogonal to } x}.$$

The claim of orthogonality is just a calculation of the inner product $\langle -Ax + \Lambda(x)x, x \rangle = 0$ □

Now we radially project this onto the sphere via

$$\pi_r : \mathbb{R}^{n+1} \setminus \{0\} \rightarrow S^n; x \mapsto \frac{x}{\|x\|}$$

the differential of this projection kills the parallel part and hence reads:

$$d\pi_r|_x(-\text{grad}_x(f)) = \frac{-Ax + \Lambda(x)x}{\|x\|}$$

If we restrict $f|_{S^n}$ we get that the gradient $-\text{grad}_y(f|_{S^n}) = -Ay + \Lambda(y)y$. To see this, let $v \in T_y S^n \subseteq T_y \mathbb{R}^{n+1}$ and remember that the inner product on S^n is induced from the euclidean one. Then

$$\begin{aligned} \langle \text{grad}_y(f|_{S^n}), y \rangle_{S^n} &= d f|_{S^n}(y) = df(y) = \langle \text{grad}_y(f), y \rangle_{\mathbb{R}^{n+1}} \\ &= \langle \text{proj}_{T_y S^n}(\text{grad}_y(f)), y \rangle_{\mathbb{R}^{n+1}} = \langle Ay - \Lambda(y)y, y \rangle_{\mathbb{R}^{n+1}} \end{aligned}$$

Here $\text{proj}_{T_y S^n}$ denotes the projection onto the subspace $T_y S^n \subseteq T_y \mathbb{R}^{n+1}$. This however tells us that the image $\widetilde{\Phi_t(x)}$ of $\Phi_t(x)$ under the radial projection solves the equation

$$\frac{d\widetilde{\Phi_t(y)}}{dt} = -\text{grad}_y(f|_{S^n})(\widetilde{\Phi_t(y)}) = -\frac{1}{2}\text{grad}_y(\Lambda|_{S^n})(\widetilde{\Phi_t(y)}) \quad \text{since } f = \frac{1}{2}\Lambda$$

Hence descending to the quotient $\mathbb{RP}^n$ yields that $\left[\widetilde{\Phi_{2t}(y)}\right]$ is a solution of the flow associated to the negative gradient of $\tilde{\Lambda}$ in $\mathbb{RP}^n$. Now the other way round should be trivial: Take a solution $\psi_t(x)$ of $\tilde{\Lambda}$ in $\mathbb{RP}^n$. Then for fixed t, x let U be a neighbourhood

of $\psi_t(x)$. This U can be lifted to S^n and the solution there solves the lifted version of the flow induced from $\tilde{\Lambda}$. And from here we can argue that the solution is part of a solution of the linear differential equation

$$\frac{d\varphi_t(x)}{dt} = -\text{grad}_{\varphi_t(x)}(f) = -Ax.$$

This concludes the proof. Hence critical points in $\mathbb{RP}^n$ are eigenvalues of A . Hence, for the function to be Morse, we need the critical points to be isolated, meaning that all eigenspaces have dimension one. So we assume that the diagonal entries in A are all different. For simplicity we assume that $\lambda_0 < \lambda_1 < \dots < \lambda_n$. Since A is diagonal we know that $[e_i]$ are the critical points.

Lemma 1.0.4. *Each critical point $[e_i]$ is non degenerate, of index i and its unstable manifold is*

$$W([e_i] \rightarrow) =: U_i = \{[x_0 : x_1 : \dots : x_i : 0 : \dots : 0] \in \mathbb{RP}^n | x_i \neq 0\},$$

whilst the stable manifold is

$$W(\rightarrow [e_i]) =: S_i = \{[0 : \dots : 0 : x_i : x_{i+1} : \dots : x_n] \in \mathbb{RP}^n | x_i \neq 0\}$$

Proof. Let B^n be the open unit ball in $\mathbb{R}^n$ and $V_i := \{[x, \dots, x_n] \in \mathbb{RP}^n | x_i \neq 0\}$. Then we define the parametrization

$$\begin{aligned} \varphi_i : B^n &\rightarrow V_i \\ (y_1, \dots, y_n) &\mapsto [y_1, \dots, y_i, \sqrt{1 - \|y\|_2^2}, y_{i+1}, \dots, y_n]. \end{aligned}$$

Now the image gives representatives of norm one and hence in this chart we calculate:

$$\begin{aligned} \tilde{\Lambda} \circ \varphi_i(y_1, \dots, y_n) &= \sum_{j=1}^i y_j^2 \lambda_{j-1} + \lambda_i (1 - \|y\|_2^2) + \sum_{j=i+1}^n y_j^2 \lambda_j \\ &= \sum_{j=1}^i y_j^2 \lambda_{j-1} + \lambda_i (1 - \sum_{j=1}^n y_j^2) + \sum_{j=i+1}^n y_j^2 \lambda_j \\ &= \sum_{j=1}^i y_j^2 (\lambda_{j-1} - \lambda_i) + \lambda_i + \sum_{j=i+1}^n y_j^2 (\lambda_j - \lambda_i) \end{aligned}$$

With this local description we can calculate the hessian in $[e_i] = \varphi_i(0)$:

$$\left. \frac{\partial^2 \tilde{\Lambda} \circ \varphi_i}{\partial y_k \partial y_l} \right|_0 = \begin{cases} 0 & \text{if } k \neq l \\ 2(\lambda_{k-1} - \lambda_i) < 0 & \text{if } k = l \leq i \\ 2(\lambda_k - \lambda_i) > 0 & \text{if } k = l > i \end{cases}$$

Hence, $[e_j]$ is a non degenerate critical point of index i . Now concerning the unstable manifolds, we pick a point $x = x_k e_k + \dots x_l e_l \in \mathbb{R}^{n+1}$ with x_k and x_l not equal to zero. This points trajectory in $\mathbb{R}^{n+1}$ under $-\text{grad}(f)$ is

$$\varphi_t(x) = x_k e^{-\lambda_k t} e_k + \dots x_l e^{-\lambda_l t} e_l.$$

Hence for t approaching negative infinity, the term $x_l e^{-\lambda_l t} e_l$ is leading, whilst for t approaching infinity, the summand $x_k e^{-\lambda_k t} e_k$ is leading. Since projectively we don't care about absolute values but rather relations having a leading coordinate meaning that relatively all other coordinates approach zero. Hence:

$$\lim_{t \rightarrow -\infty} ([\varphi_t(x)]) = [e_l] \quad \text{and,} \quad \lim_{t \rightarrow +\infty} ([\varphi_t(x)]) = [e_k].$$

□

Definition 1.0.5 (Orientations). For orienting the unstable manifolds we make use of the projection $\mathbb{R}^{n+1} \supseteq S^n \rightarrow \mathbb{RP}^n$. Hence, we can lift $T_{[e_i]}\mathbb{RP}^n$ to $T_{\pm e_i}S^n$ and include that into $T_{\pm e_i}\mathbb{R}^{n+1}$, where we have the maps $Q_{\pm e_i} : \mathbb{R}^{n+1} \rightarrow T_{\pm e_i}\mathbb{R}^{n+1}$. In this way we get two canonical lifts of the tangent space:

$$\begin{array}{ccc} T_{[e_i]}\mathbb{RP}^n & \xrightarrow{d\pi_{e_i}^{-1}} & T_{e_i}S^n \\ \downarrow d\pi_{-e_i}^{-1} & & \downarrow Q_{e_i}^{-1} \\ T_{-e_i}S^n & \xrightarrow{Q_{-e_i}} & \langle (e_j)_{j \neq i} \rangle \end{array}$$

In general, the diagram is not commutative. To see this notice how $\pi \circ -\text{id} = \pi$. Hence $d\pi|_{-e_i} \circ d - \text{id}|_{e_i} = d\pi|_{e_i}$ and with that:

$$\begin{aligned} Q_{-e_i}^{-1} \circ d\pi|_{-e_i}^{-1} \circ d\pi|_{e_i} \circ Q_{e_i} &= Q_{-e_i}^{-1} \circ d\pi|_{-e_i}^{-1} \circ d\pi|_{-e_i} \circ d(-\text{id})|_{e_i} \circ Q_{e_i} \\ &= Q_{-e_i}^{-1} \circ d\pi|_{-e_i}^{-1} \circ d\pi|_{-e_i} \circ Q_{-e_i} \circ -\text{id} \\ &= Q_{-e_i}^{-1} \circ Q_{-e_i} - \text{id} = -\text{id}. \end{aligned}$$

Hence, we can define a canonical orientation on $T_{[e_i]}\mathbb{RP}^n$ that is given by $(e_j)_{j \neq i}$ or equivalent $(-e_j)_{j \neq i}$ depending on the chosen lift. Now in the same way we can induce orientations to the unstable manifolds U_i by two equivalent ways:

$$\begin{array}{ccc} T_{[e_i]}U_i & \hookrightarrow & T_{[e_i]}\mathbb{RP}^n \\ \uparrow & & \uparrow \\ \langle (e_0, \dots, e_{i-1}) \rangle & \hookrightarrow & T_{e_i}S^n \end{array} \quad \text{or equivalently} \quad \begin{array}{ccc} T_{[e_i]}U_i & \hookrightarrow & T_{[e_i]}\mathbb{RP}^n \\ \uparrow & & \uparrow \\ \langle -e_0, \dots, -e_{i-1} \rangle & \hookrightarrow & T_{-e_i}S^n \end{array}$$

Now we know how the groups in the Morse chain complex are looking. The next step is the boundary operator:

Lemma 1.0.6. *The differential in $C_*(\mathbb{RP}^n, \Lambda^*)$ with the right orientations act as*

$$\partial_i([e_i]) = (-1)^{i-1}[e_{i-1}] - [e_{i-1}].$$

Proof. First we notice that $W([e_i] \rightarrow [e_{i-1}])$ is a one dimensional projective space with two points missing. If we would look at the Morse homology with coefficients in $\mathbb{Z}_2$ we would be done here, since $\mathbb{RP}^1 \setminus \{p, q\}$ consists of two lines. Hence, over $\mathbb{Z}_2$ the boundary operator is trivial. However we want to use the orientations from above. Now let

$$x \in W([x_i] \rightarrow [e_{i-1}]) = \{[0 : \dots : 0 : x_{i-1} : x_i : 0 : \dots : 0] \mid x_{i-1} \neq 0 \text{ and } x_i \neq 0\}.$$

Then we can choose representatives, such that $x_{i-1}^2 + x_i^2 = 1$, and (x_{i-1}, x_i) is in the upper half plane meaning that $x_i > 0$. Now take a $x \in W([e_i] \rightarrow [e_{i-1}])$. Then $T_x W([e_i] \rightarrow [e_{i-1}])$ has an oriented basis given as the pushforward of $v = x_i e_{i-1} - x_{i-1} e_i$ under the projection $T_x S^n \rightarrow T_x \mathbb{RP}^n$. Now we want to analyze how this relates to the orientation induced by the flow and the two unstable orientations: if both x_i and x_{i-1} are positive, then v points in the direction of the flow. Else, if x_{i-1} is negative, $-v$ points in the direction of the flow. The parallel transport of $T_{[x_{i-1}]} U_{i-1}$ to the normal space of $-\text{grad}_x(\tilde{\Lambda})$ induced by the negative flow maps the basis $e(0, \dots, e_{i-2})$ of $T_{[e_{i-1}]} U_i$ to $(e_0, \dots, e_{i-2})$ in the orthogonal complement of $-\text{grad}_c(\tilde{\Lambda})$ in $T_x \mathbb{RP}^n$. So we need to compare the ordered basis $(v, e_0, \dots, e_{i-2})$ to the transported ordered basis $(e_0, \dots, e_{i-1})$ that started in $T_{[e_i]} U_i$. This procedure maps the basis vectors as follows:

$$e_j \mapsto \begin{cases} x_i e_{i-1} - x_{i-1} e_i & \text{if } j = i-1 \\ e_j & \text{else.} \end{cases}$$

In other words, we need to ask when the two basis

$$((x_i e_{i-1} - x_{i-1} e_i), e_0, \dots, e_{i-2}) \quad \text{and} \quad (e_0, \dots, e_{i-2}, x_i e_{i-1} - x_{i-1} e_i),$$

if $x_{i-1} > 0$ are comparable. Then, the basis carry the same orientation, if and only if i is odd, since then it takes evenly many transpositions of coordinates.

If $x_{i-1} < 0$ we notice, that our representation of $[e_{i-1}]$ is given by $-e_{i-1}$ and hence the unstable tangend space has the basis $-e_0, \dots, -e_{i-2}$. Hence in x we get the oriented basis $(-v, -e_0, \dots, -e_{i-2})$. We compare this basis to basis that comes from transporting the basis $(e_0, \dots, e_{i-1})$ of $T_{[e_i]} U_{[e_i]}$ to x resulting in the basis $(e_0, \dots, e_{i-2}, v)$. By definition we would now move v to the front and change its sign, whilst not changing the orientation. This results in th basis $(-1)^{i-2+1}(-v, e_0, \dots, e_{i-2})$. The sign comes from the number of needed permutations plus one change of sign. Now we remove $-v$ and compare the basis

$$(-e_0, \dots, -e_{i-2}) \quad \text{and} \quad (-1)^{i-2+1}(-v, e_0, \dots, e_{i-2}) = (-1)^{i-2+1+i-2}(-v, -e_0, \dots, -e_{i-2}).$$

Since the sign of $(-1)^{i-2+1+i-2}$ is always negative, they never agree. Hence, the Morse boundary operator counts minus two flow lines, if i is even and else, the two flow lines cancel out concluding the proof. $\square$

why is the half circle a truthfully depiction and model?

Now for odd n , EM_1 of the form:

$$0 \quad \langle [e_0] \rangle \xrightarrow{-2} \langle [e_1] \rangle \xrightarrow{0} \langle [e_2] \rangle \xrightarrow{-2} \dots \xrightarrow{-2} \langle [e_n] \rangle \xrightarrow{0} 0$$

and for even n :

$$0 \quad \langle [e_0] \rangle \xrightarrow{-2} \langle [e_1] \rangle \xrightarrow{0} \langle [e_2] \rangle \xrightarrow{-2} \dots \xrightarrow{0} \langle [e_n] \rangle \xrightarrow{0} 0$$

Hence we have the EM_2 pages as follows:

$$0 \quad \langle [e_0] \rangle \quad 0 \frac{\langle [e_1] \rangle}{\langle -2[e_1] \rangle} \quad 0 \quad \frac{\langle [e_3] \rangle}{\langle -2[e_3] \rangle} \quad \dots \quad \frac{\langle [e_n] \rangle}{\langle -2[e_n] \rangle} \quad 0$$

if n is odd, and in the even case:

$$0 \quad \langle [e_0] \rangle \quad \frac{\langle [e_1] \rangle}{\langle -2[e_1] \rangle} \quad 0 \quad \frac{\langle [e_3] \rangle}{\langle -2[e_3] \rangle} \quad \dots \quad \langle [e_n] \rangle \quad 0$$

When it comes to higher differentials, we notice the following: Only for odd r the differentials are possible nontrivial. Hence, the differentials change the parity of p . Hence the only possible way this is not trivial is, if d_r maps to $p = 0$. Hence for $r = 3$ the only differential we need to look at is

$$d_3^3 : \frac{\langle [e_3] \rangle}{\langle -2[e_3] \rangle} \rightarrow \langle [e_0] \rangle$$

To understand this differential we need to get specific about the used filtration. We define the filtration inductively as follows: For any i define $c_i \in (\lambda_i, \lambda_{i+1})$ and with this we define the filtration

$$N_{-1} := \emptyset, \quad N_i = \tilde{\Lambda}(-\infty, c_i) \text{ the sublevel sets, } \quad N_n := \mathbb{RP}^n.$$

With this filtration it is somewhat easy to see that (N_i, N_{i-1}) gives a regular index pair of $[e_i]$. One just needs to go through the definitions: First it is clear, that the only critical point inside of the closure of $N_i \setminus N_{i-1}$ is $[e_i]$ and it is in the interior. Secondly, the positiv lime invariance is clear, since $\tilde{\Lambda}$ decreases along flow lines. And N_{i-1} is an exit set also by definition. Finally the cofibration property of the pair N_i, N_{i-1} is to be shown. For this we notice, that c_i is a regular value of R . Hence there exists an $\varepsilon > 0$, such that $f^{-1}([a, a + \varepsilon))$ contains no kritical point and hence this defines an open color of N_i inside N_{i+1} . This open color can easily be shown to be an NDR pair, which in turn is a cofibration. Those are standart arguments seen in [Whitehead1978] Chapter 5.1.

Lemma 1.0.7. *In the filtration, each N_i is homotopically equivalent to $\mathbb{RP}^i$.*

Proof. Notice how for all i the closure $\overline{U_i}$ is homotopically equivalent to $\mathbb{RP}^i$. So we want to show that $N_i \simeq \overline{U_i}$ and for this we start by noticing that $N_i \simeq \tilde{\Lambda}^{-1}(-\infty, \lambda_{i-1} + \varepsilon)$ for all $\varepsilon > 0$ small enough. Now the next step is to get rid of epsilon. Let L_ε denote $\tilde{\Lambda}^{-1}(-\infty, \lambda_{i-1} + \varepsilon)$ and hence we want to show that $L_\varepsilon \simeq L_0$. For this we define

$$H : L_\varepsilon \times [0, 1] \rightarrow L_\varepsilon \quad ([x], t) \text{ with } \|x\| = 1 \mapsto \left[\sum_{j=1}^i x_j^2 e_j + (1-t) \sum_{j>i} x_j^2 e_j \right]$$

First we need to check if this map is well defined in two senses. Is the definition invariant under the chosen representative and do we map into L_ε . For the independence of the chosen representative we notice that the image of x and $-x$ differ by sign and hence define the same equivalence class. Also applying $\tilde{\Lambda}$ to the image we notice that the image is less than $\lambda_i + \varepsilon$. To see this we normalize the image of H_t for any t and apply $\tilde{\Lambda}$:

$$\tilde{\Lambda}(H_t([x])) = \frac{\sum_{j \leq i} \lambda_j x_j^2 + (1-t) \sum_{j > i} \lambda_j x_j^2}{\sum_{j \leq i} x_j^2 + (1-t) \sum_{j > i} x_j^2}$$

□

Now that we have our filtration we can start looking at higher differentials: For this we start with $n = 3$ and then go up. In the case of $n = 3$ we are trivial, since d_3^0 is given by the composition

$$\begin{array}{ccccc} & & & & K^0(N_3) \\ & & & & \downarrow i^* \\ K^0(N_0, \emptyset) & \xrightarrow{\gamma} & K^0(N_0) & \xrightarrow{\widetilde{\alpha^{-2}}} & K^0(N_2) \\ & \searrow d_3 & & & \downarrow \delta \\ & & & & K^1(N_3, N_2) \end{array}$$

Here α^{-1} is any left inverse of α^2 or taking the preimage (the well definedness or rather independence of the chosen inverse was shown in the definition of the derived exact couple). The vertical sequence is exact and since $N_2 = N_3 \setminus D$ where D is contractible, we have that i^* is surjective and hence δ is trivial and thereby $dM_3^0 = 0$. The same arguments yields that for $\mathbb{RP}^4$ the map $d_3^1 = 0$. And hence for $\mathbb{RP}^n$ with $n \leq 4$ all higher differentials are trivial. Let $n = 5$. Then we have a chance for a nontrivial d_3^0 . Again, the map is given via the diagram

$$\begin{array}{ccccc} & & & & K^0(N_3) \\ & & & & \downarrow i^* \\ K^0(N_0, \emptyset) & \xrightarrow{\gamma} & K^0(N_0) & \xrightarrow{\widetilde{\alpha^{-2}}} & K^0(N_2) \\ & \searrow d_3 & & & \downarrow \delta \\ & & & & K^1(N_3, N_2) \end{array}$$

However this time, N_2 is not just N_3 with something collapsable removed and hence there is no clear argument why i^* should be surjective.